Date : Nov 17 , 2025

Team Member : Ritika

Task 01 : Descriptive Analytics Case Study

Scenario

You are provided a dummy dataset from Super Irvin, a supermarket located in LA, California . The dataset contains sales information of a particular item: "Organic Almond Milk".

Your job is to perform Descriptive Analytics and understand the supermarket sales pattern.

PART A — Identify Data Types

Identify for each column:
(Nominal / Ordinal / Interval / Ratio / Quantitative / Qualitative)

Column Name	Data Type (To be answered by Ritika)
Transaction_ID	?
Customer_Age	?
Gender	?
Purchase_Date	?
Units_Sold	?
Price_per_Unit	?
Total_Sales	?
Customer_Type	?
Satisfaction_Rating	?

PART B — Statistical Analysis Table

Ritika, create a table containing:

- Count
- Mean
- Median
- Mode
- Minimum
- Maximum
- Standard Deviation
- Percentiles (25%, 50%, 75%)

For these numeric columns:

1. Customer_Age
2. Units_Sold
3. Total_Sales
4. Satisfaction_Rating

PART C — Summary Writing Task

- What the statistics show
- Trends in customer age, purchases, and satisfaction
- Which customer type buys more
- Insights that Super Irvin Mall can use
- Any pattern in satisfaction rating

Visualization Task

Create the following 4 charts (Google Colab -Python):

1. **Bar Chart** — Gender vs Units Sold
2. **Line Chart** — Daily Total Sales
3. **Histogram** — Customer Age
4. **Pie Chart** — Customer Type Distribution

For any assistance, you can mail me.
Thank you.